

Introduction

Intensity Transformations

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Agenda

Intensity Transformations

Histograms and Histogram Equalization

Morphological Image Processing

Introduction to Python and OpenCV

Image Processing

Computer Graphics

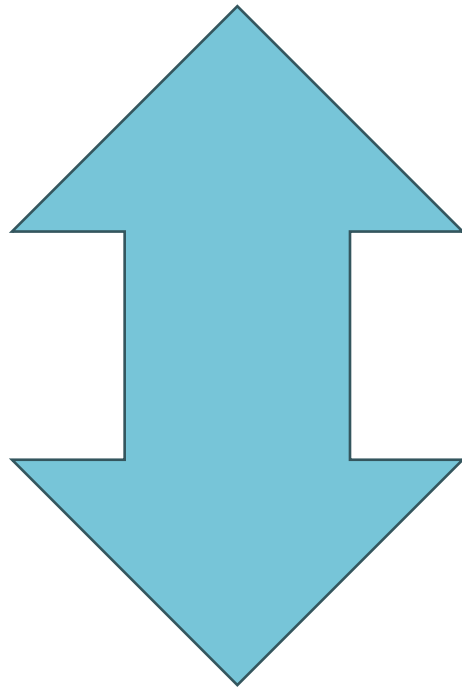
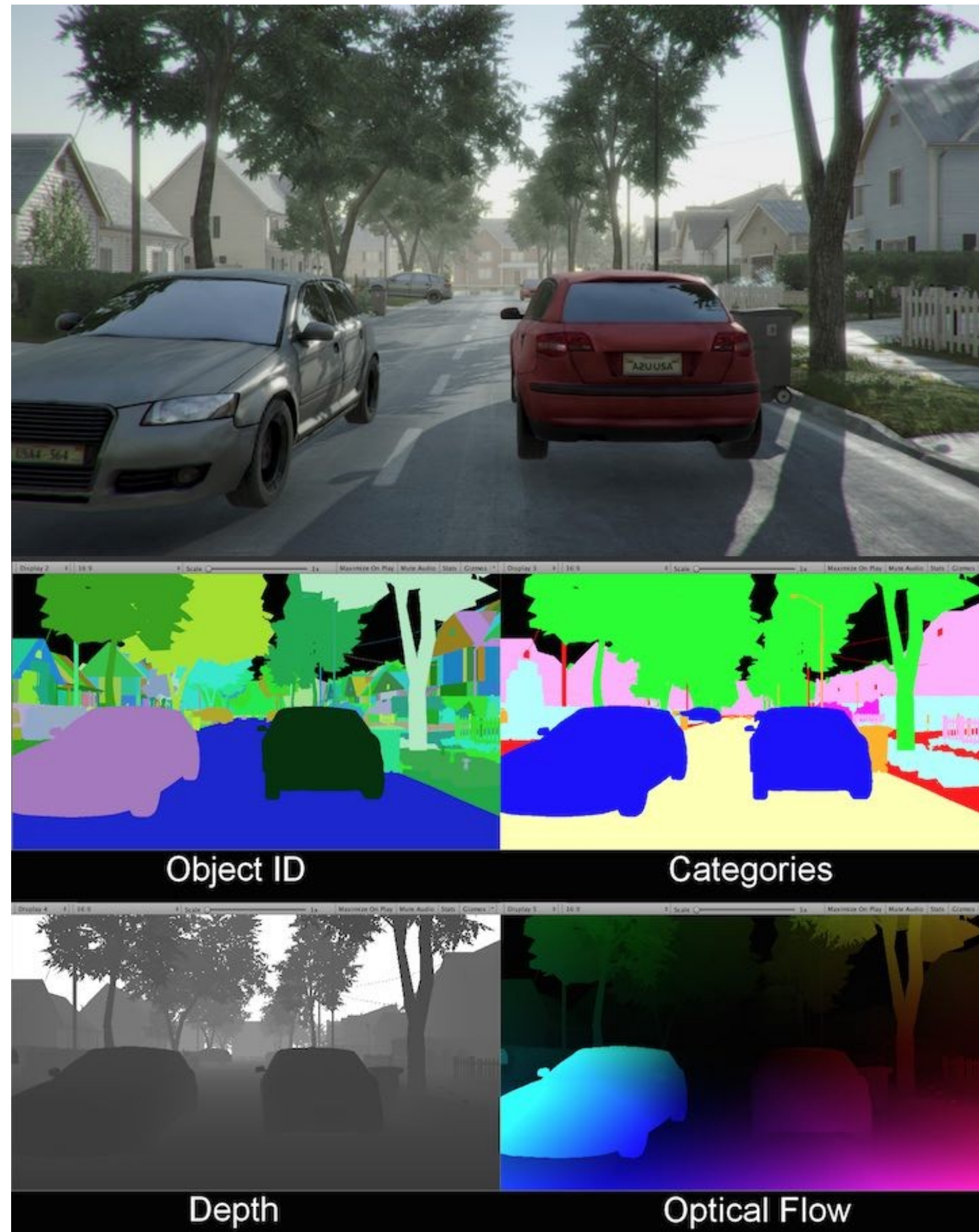


Image Processing

Generate realistic images from objects and scenes

Recognize objects and scenes from images

Graphics from Computer Vision



Creating realistic images in Unity

Training data for Machine Learning Applications

Different levels of image processing

High Level Vision

Computer Vision

Interpreting images

Recognizing objects

Mid Level Vision

Image Processing

Image enhancement

(contrast correction, noise reduction, etc.)

Recognizing low level features

(edges, lines, etc.)

Low Level Vision

What is an image?

Image Preprocessing

Strengthen information

(for example scale codomain until structures of interest will appear)

Reduce perturbation

(noise, artifacts, illumination errors, jitter, etc.)



Image processing methods

Distinguish between

- Single pixel operations
Operations depend on single points,
i. e. only the grey value or color value will be transformed
- Neighborhood operations
The result of an operation depends on the pixel together with its neighborhood
(for example 3x3 pixels), filters, etc.

Intensity Transformations

Changing intensity values using a mathematical function

Examples:

- Thresholding
- Changing brightness
- Changing contrast
- Gamma correction

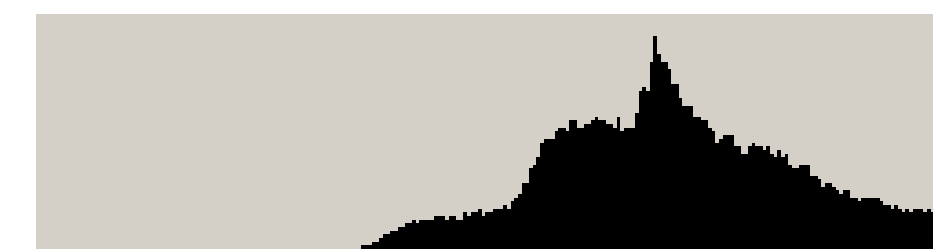
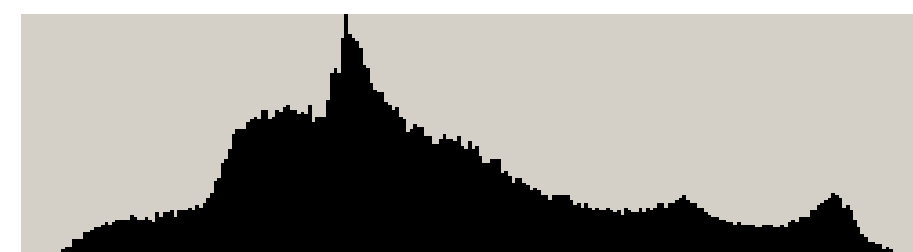
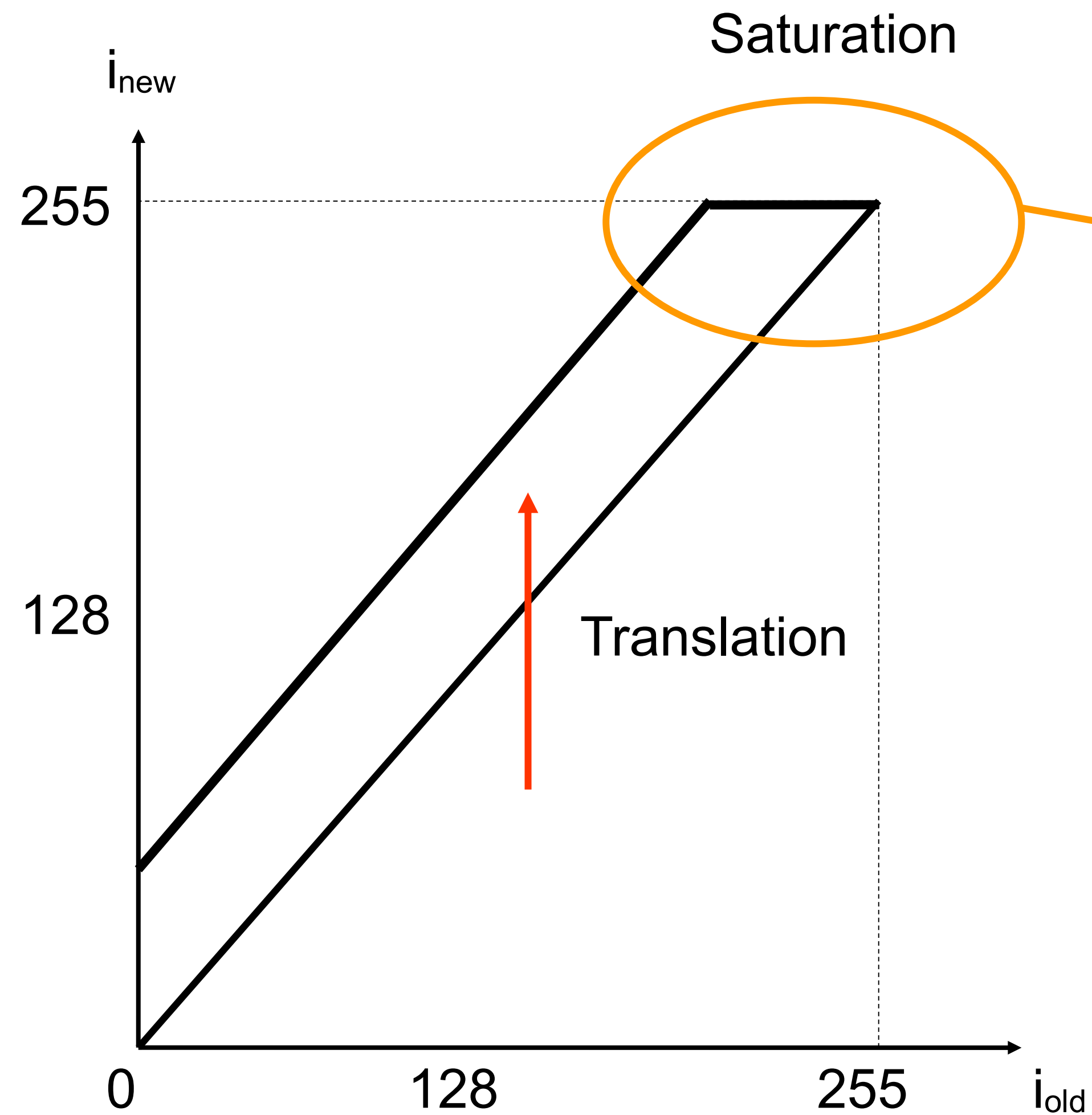
$$i_{\text{new}} = f(i_{\text{old}})$$

Intensity Transformations

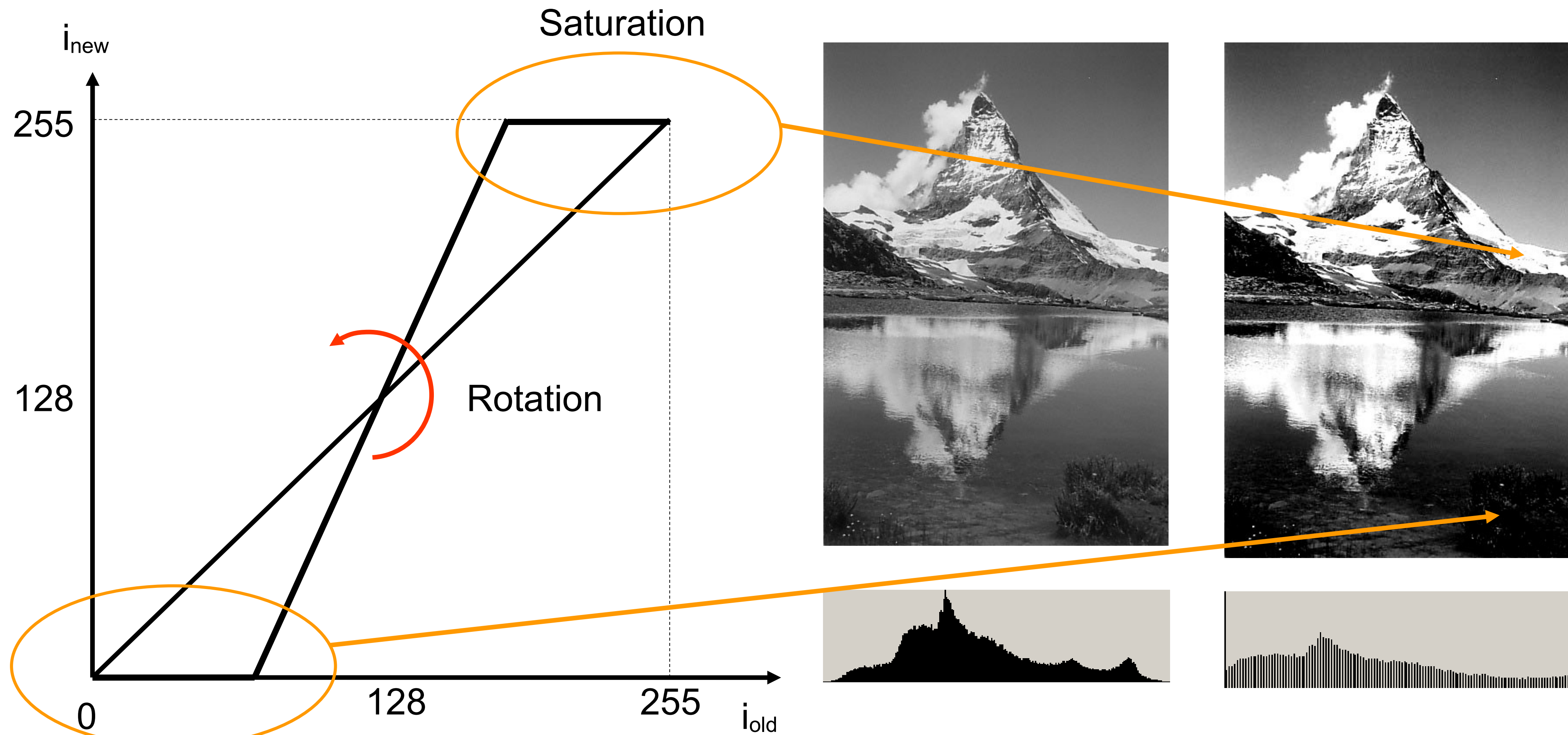
How do the functions of following single pixel transformations look like?

- Change in brightness
- Contrast enhancement
- Autoscaling
- Gamma correction

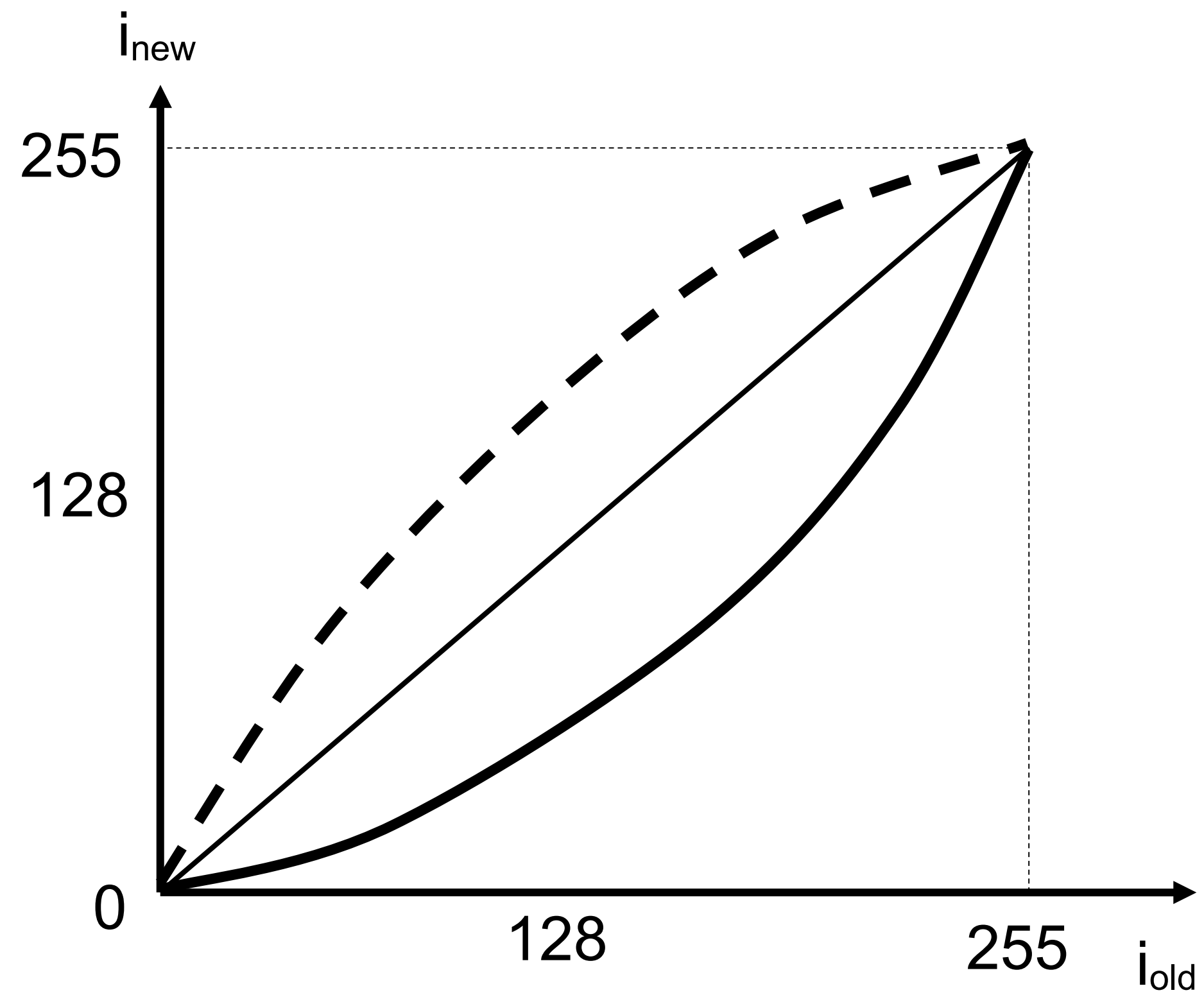
Changing Brightness



Changing Contrast



Gamma correction



$$f(i) = i^\gamma$$

Histogram

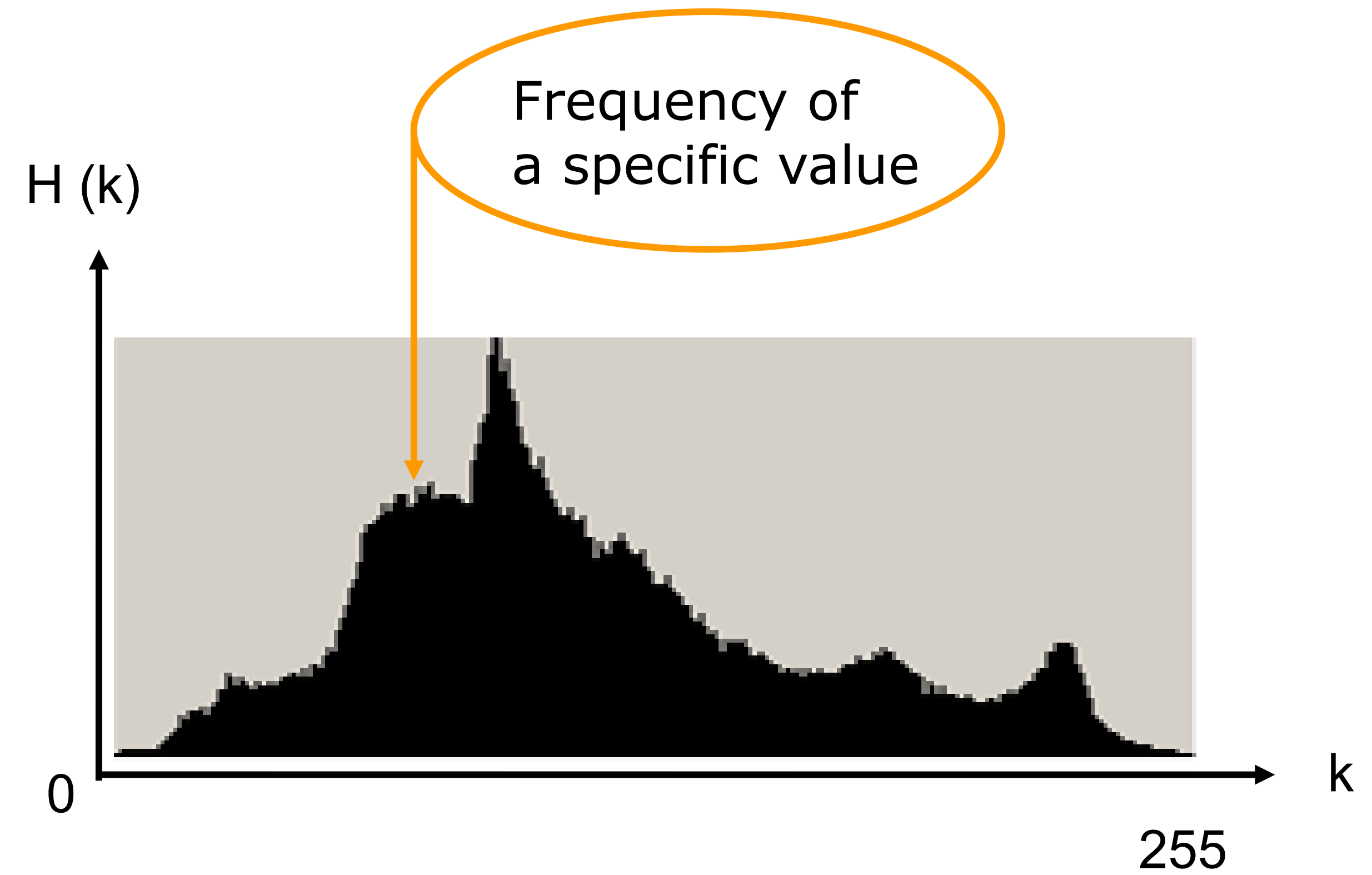
Color distribution: How often does a color / intensity appear in an image?

$$H(k) = \left| \left\{ x_i \mid v_k \leq I(x_i) < v_{k+1} \right\} \right|$$

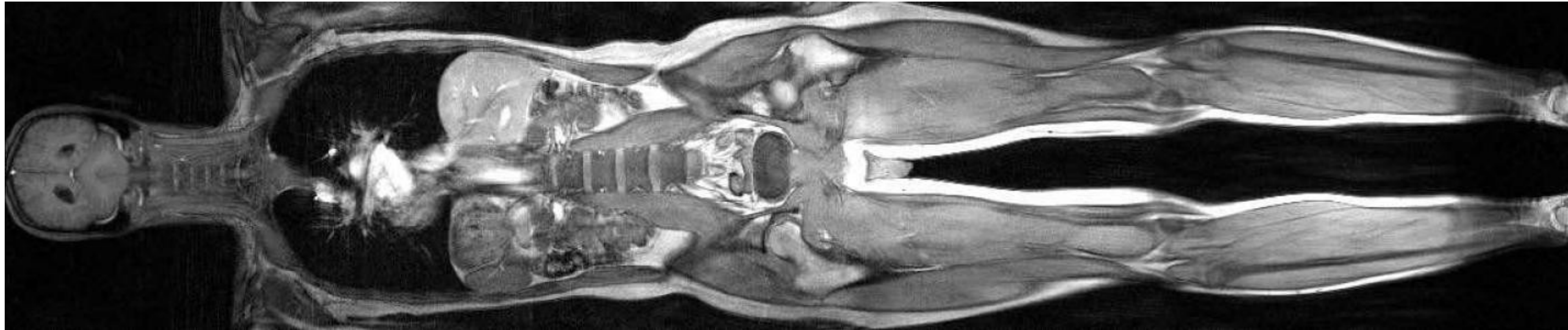
k: Number of bins in histogram

v_k : Intensity of bins

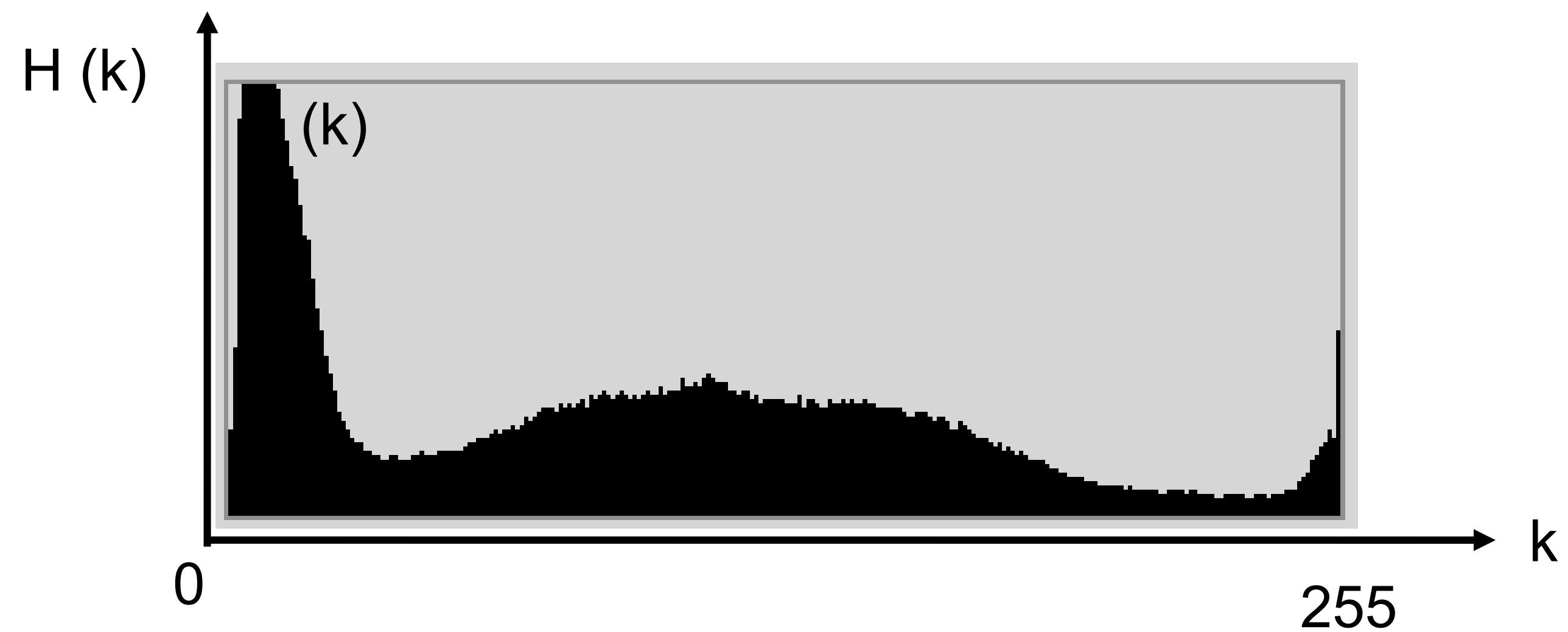
Histogram example



Histogram example



CT scan: Visible human project



Pixel Operations

- Pixel operations change intensity or color

$$i_{\text{new}} = f(i_{\text{old}})$$

- How is the histogram changed?

$$H_{\text{new}}(i) = H_{\text{old}}(i) \cdot \frac{1}{f'(i)}$$

Histogram Equalization

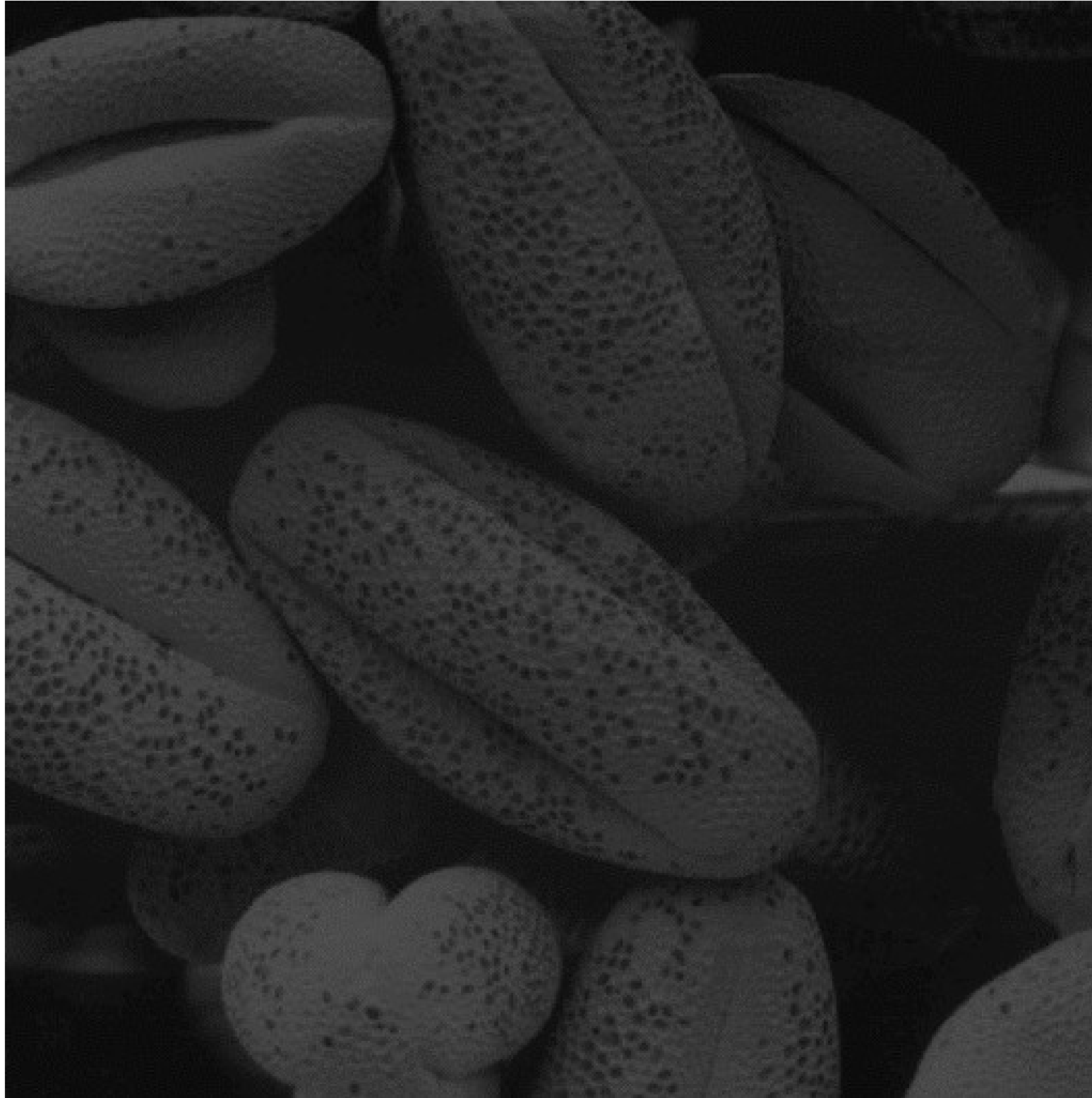
- Objective:

Transformation of the gray values

so that the histogram of the new image is as balanced as possible

- Auto-Contrast Function in image processing tools

Histogram Equalization: Example



Original Image

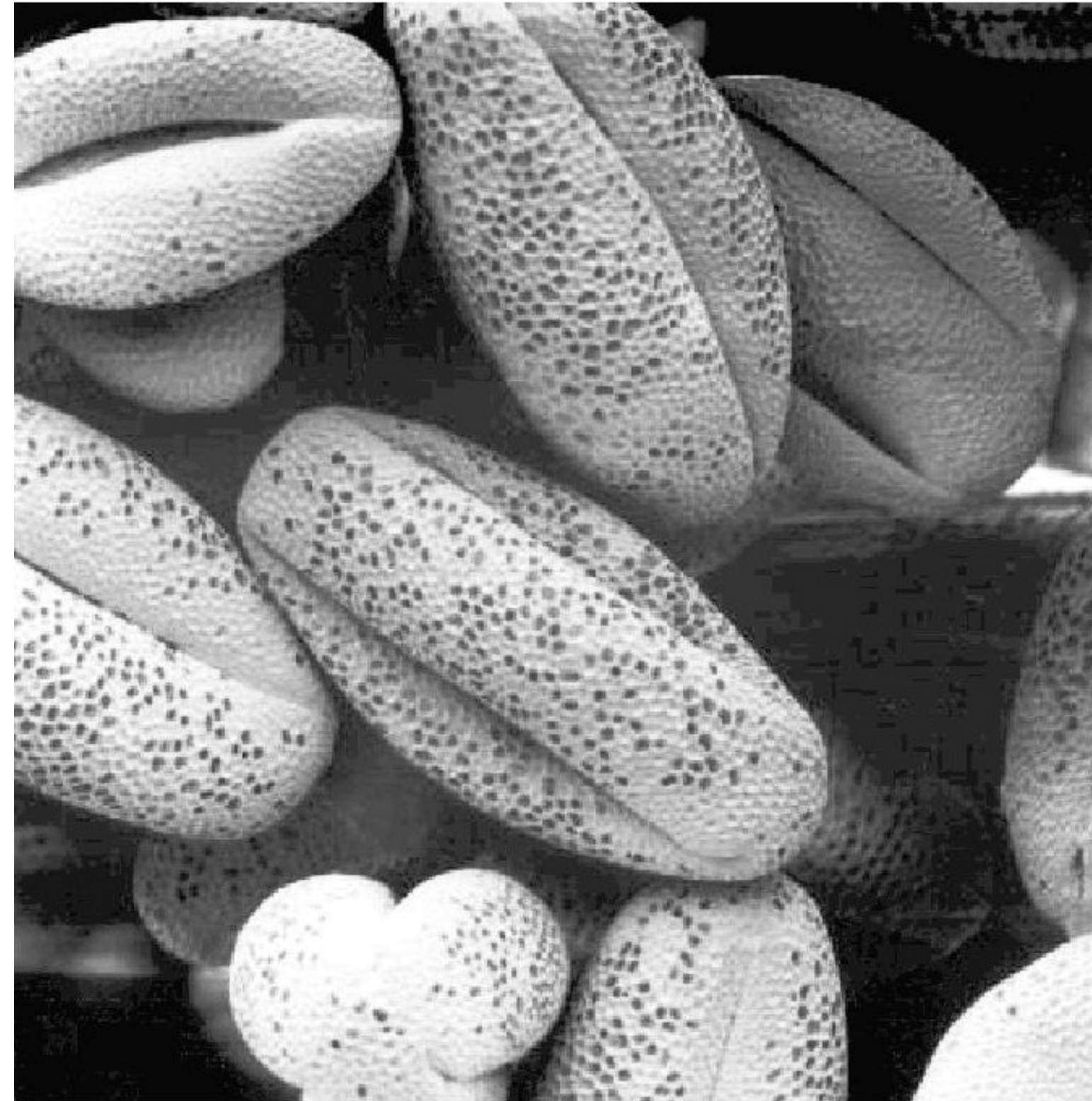


Image after histogram equalization

Histogram Equalization: Computation

If new intensity frequencies are to be constant

$$H_{\text{new}}(i) = c = \text{const.}$$

we get

$$f'(i) = \frac{1}{c} H_{\text{old}}(i)$$
$$f(i) = \frac{1}{c} \int_0^i H_{\text{old}}(j) dj$$

Histogram Equalization: Example

What does the histogram equalization look like for the following 'Image':

0	1	1	2	2
2	1	1	2	2
4	5	6	7	7
8	8	7	7	7
8	8	9	9	9

Morphological Image Processing

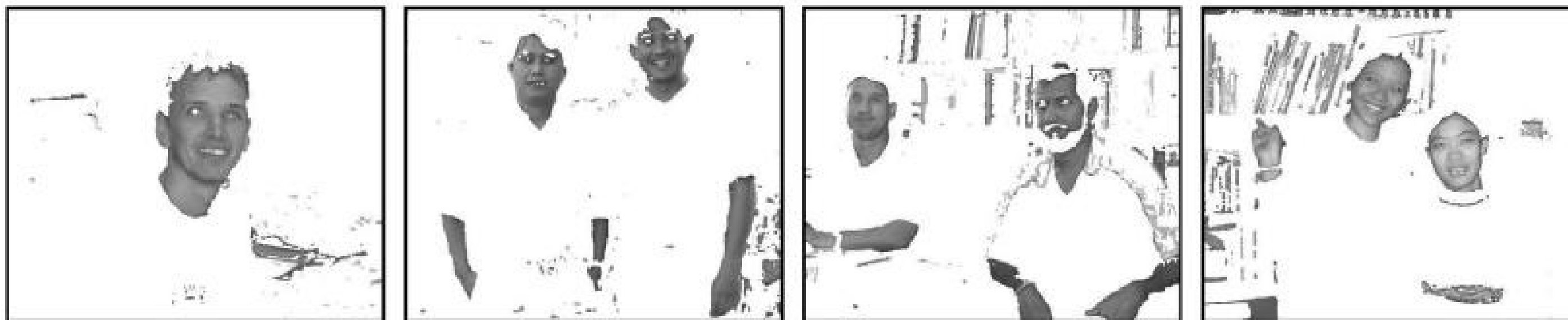
- Morphological operations are used to process binary images
- Binary images can be generated, for example, by setting a threshold value
- However, the resulting regions often require further processing.

Morphological Image Processing: Example

- Compute regions by skin detector and threshold value
- How many faces can be seen in the image?



(a)



Morphological Image Processing: Set Theory

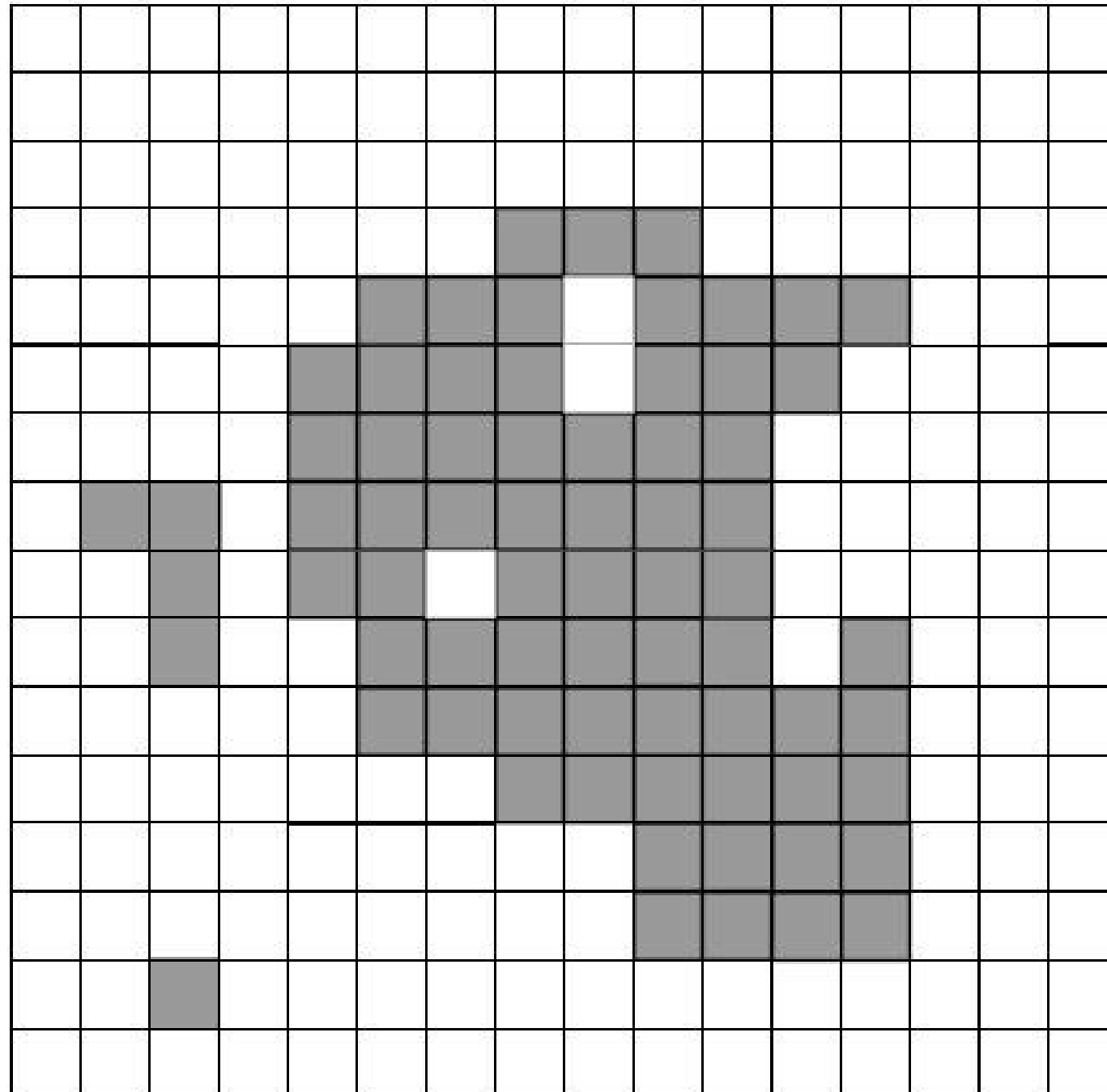
- Morphological operations can be represented by set operations
- The image is combined with a structural element, which is defined as a matrix, for example

$$S = \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

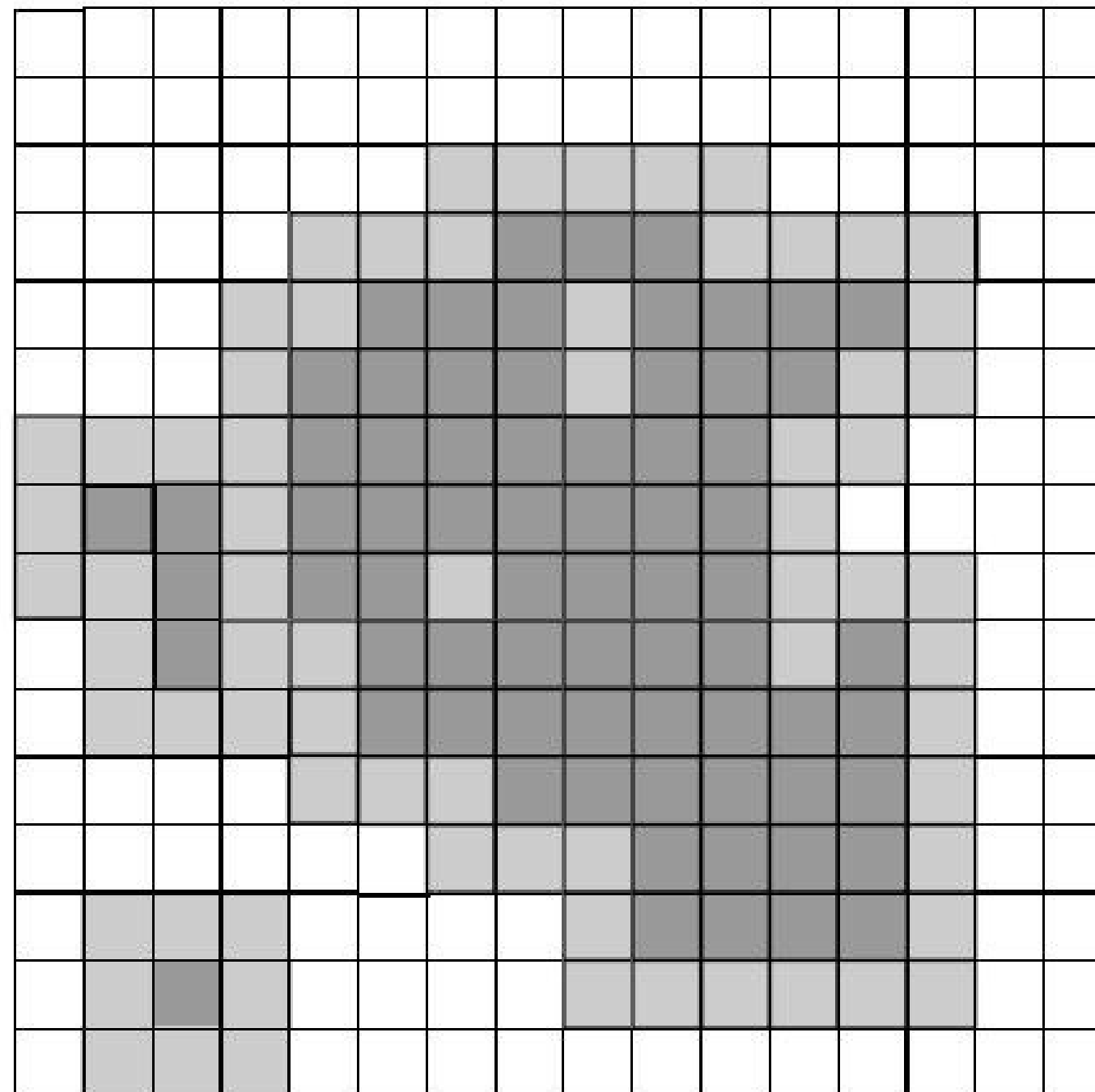
Morphological Image Processing: Dilate

$$A \oplus S = \{x \mid A \cap S_x \neq \emptyset\}$$

a



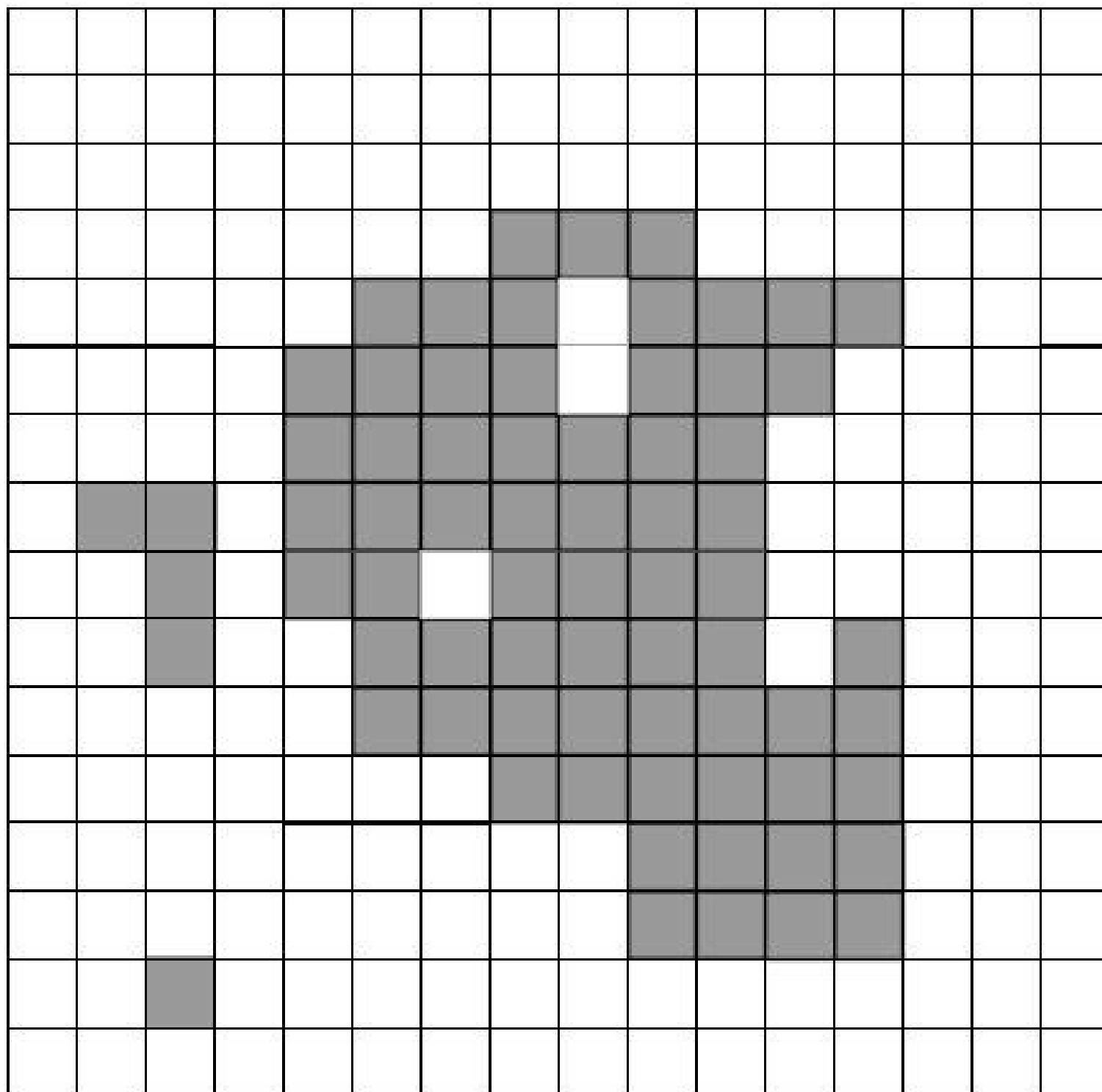
b



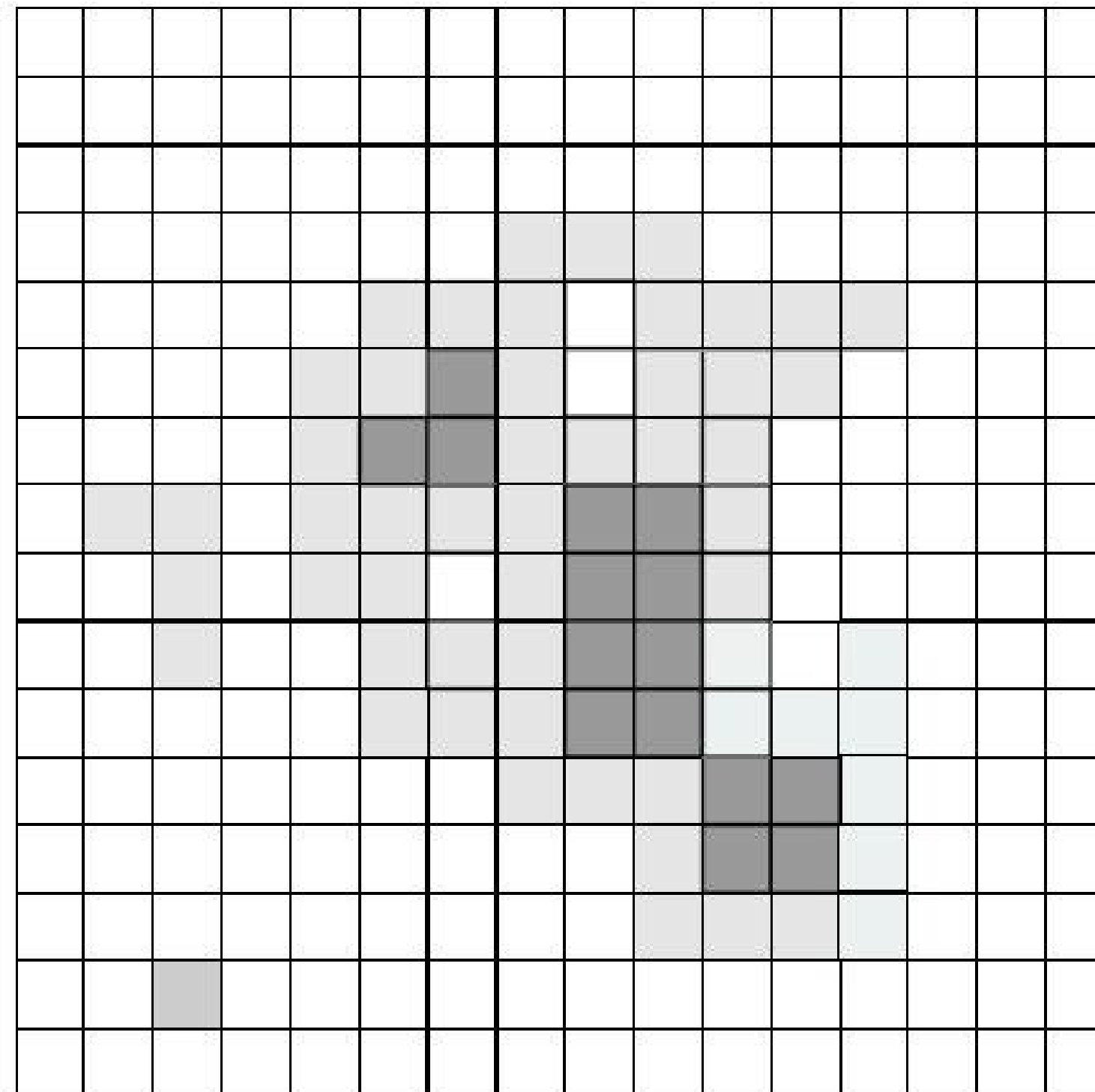
Morphological Image Processing: Erode

$$A \ominus S_x = \{x | \bar{A} \cap S_x = \emptyset\} = \{x | A \supseteq S_x\}$$

a



c



Morphological operations

- Dilate: Thickens the object
- Erode: Thins the object
- Closing: Dilate followed by Erode, closes the object (eliminates holes)
- Opening: Erode followed by Dilate, opens (separates objects)

Opening / Closing: Example



Connected Component Labeling

- How many objects can be seen on a (binary) image?
- How “large” (number of pixels) are the objects?
- Which regions are contiguous?

Connected Component Labeling: 4 / 8 neighborhood

A set of pixels can be connected to each other via a 4-neighborhood or a 8-neighborhood

	n	
n	p	n
	n	

4-neighborhood

n	n	n
n	p	n
n	n	n

8-neighborhood

Connected Component Labeling: Example

A set of pixels can be connected to each other via a 4-neighborhood or a 8-neighborhood

1	1	1	0	0	0	0	0
1	1	1	0	1	1	0	0
1	1	1	0	1	1	0	0
1	1	1	0	0	0	1	0
1	1	1	0	0	0	1	0
1	1	1	0	0	0	1	0
1	1	1	0	0	1	0	0
1	1	1	0	0	0	0	0

Original image (binary)

1	1	1	0	0	0	0	0
1	1	1	0	2	2	0	0
1	1	1	0	2	2	0	0
1	1	1	0	0	0	3	0
1	1	1	0	0	0	3	0
1	1	1	0	0	0	3	0
1	1	1	0	0	4	0	0
1	1	1	0	0	0	0	0

Detected regions with labels

Exercise: Count Spheres



OpenCV

- Image Processing Library
- Originally from Intel 1999
- Cross-Platform, Open Source, Optimized
- Written in C/C++, interface also in Python and other languages
- It is a library of functions:
You still need to know what the purpose of the functions is.

